Motor Driver — Breadboard Layout

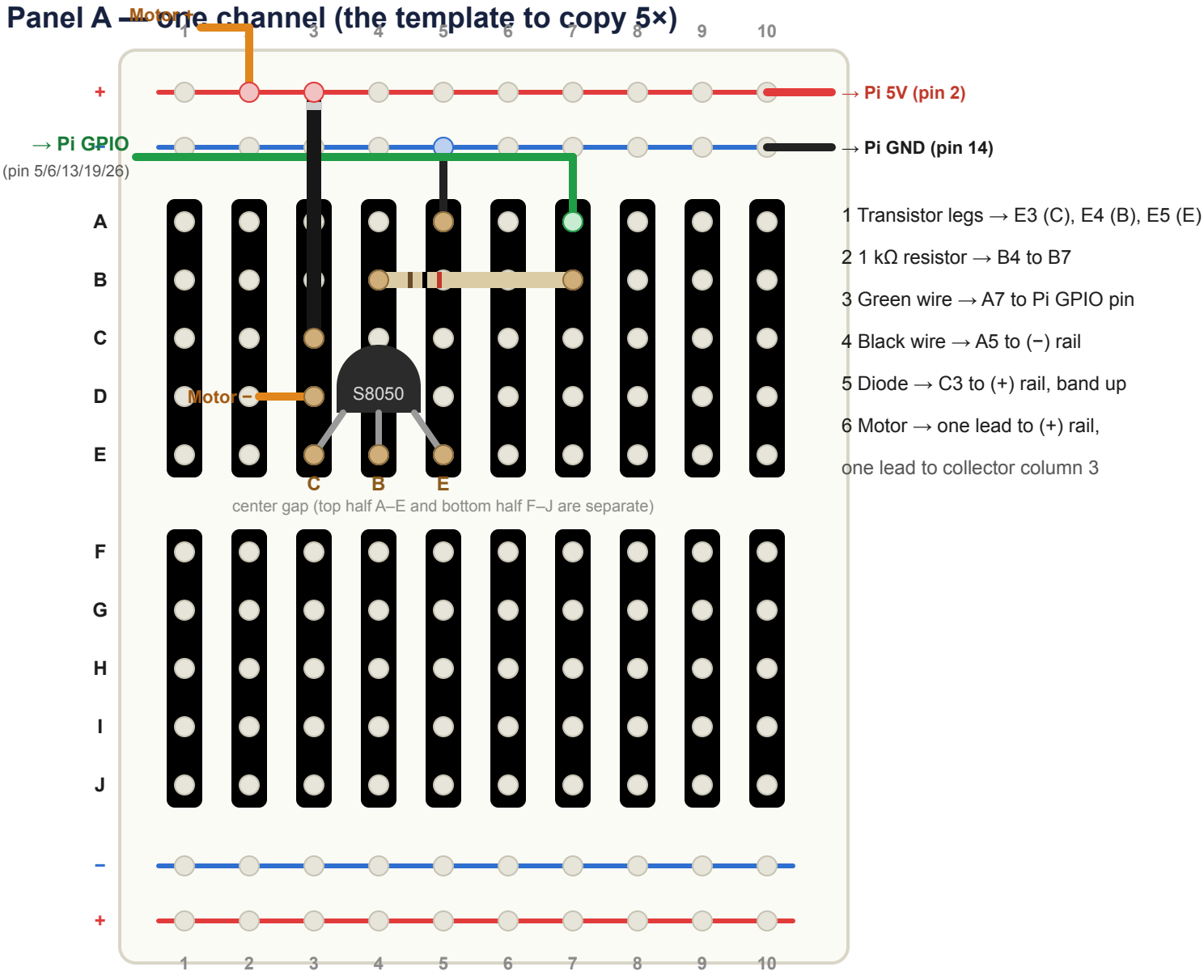
Copy Panel A once per motor (build all 5). Holes shown at real spacing.

PARTS & WIRES

NPN transistor (S8050/2N222) · 1 kΩ resistor · 1N4007 diode (band = stripe)

+5 V · GND · GPIO signal · motor leads

Panel A — One channel (the template to copy 5×)



1 Transistor legs → E3 (C), E4 (B), E5 (E)

2 1 kΩ resistor → B4 to B7

3 Green wire → A7 to Pi GPIO pin

4 Black wire → A5 to (−) rail

5 Diode → C3 to (+) rail, band up

6 Motor → one lead to (+) rail,

one lead to collector column 3

CHECK THE TRANSISTOR PINOUT

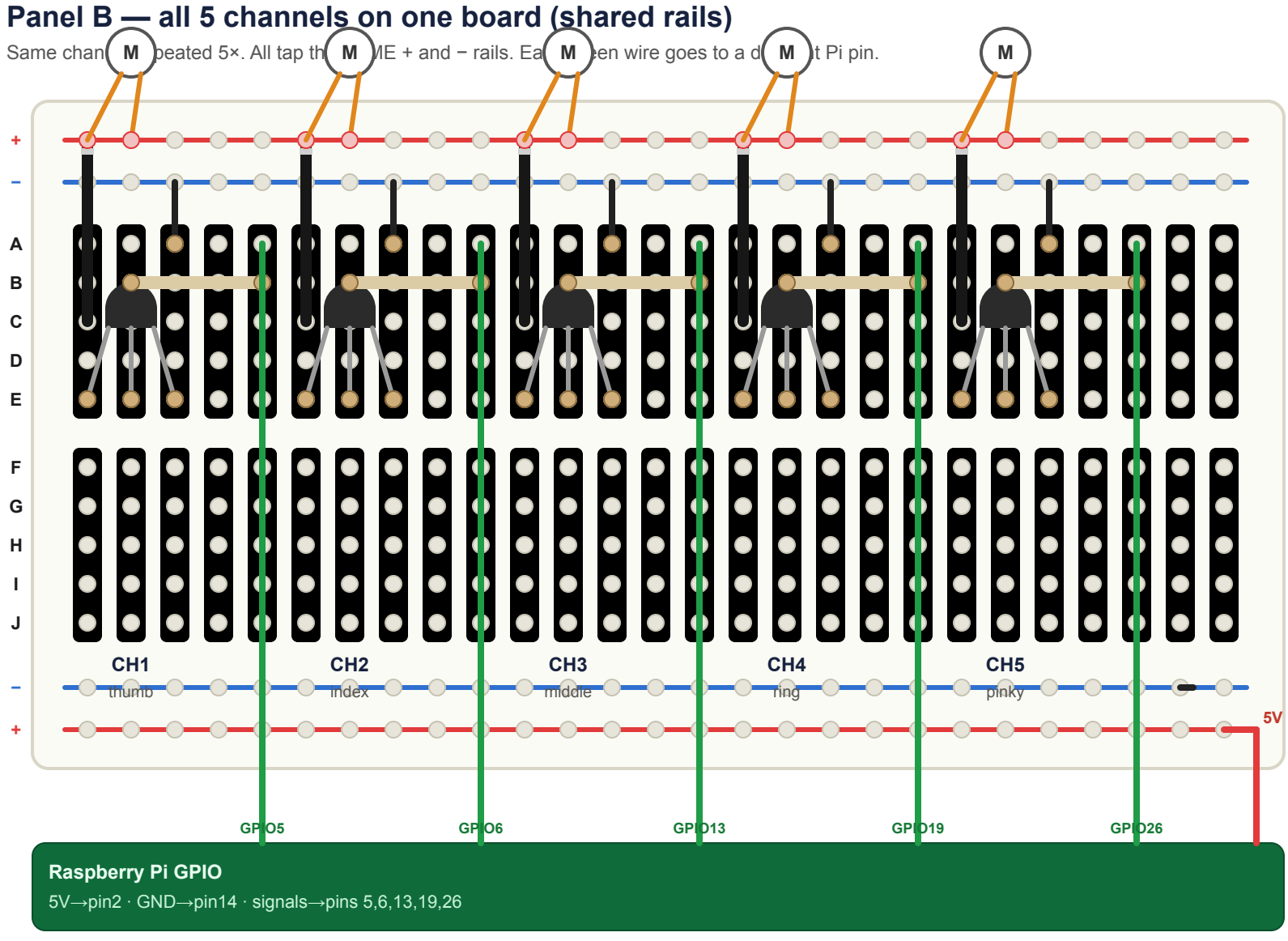
Leg order C–B–E differs by part. Orient YOUR transistor so its legs land in the C / B / E columns shown above.

MOTOR VOLTAGE

If your coin motors are 3 V-rated, add a series resistor or use a 3 V feed — don't run 5 V continuously.

Panel B — all 5 channels on one board (shared rails)

Same chan[illegible]peated 5×. All tap th[illegible]ME + and − rails. Ea[illegible]een wire goes to a d[illegible]t Pi pin.



HOW TO READ THIS

- Holes in the same column (A–E together, or F–J together) are electrically joined — that's how legs connect without wires.
- The + and − rails run the whole length; every channel taps the same two rails. Feed them once from the Pi (pin 2 and pin 14).
- Build ONE channel from Panel A, test it, then repeat four more times shifting right. Motor leads and Pi wires run off-board.